Journal of Translational Medicine

RESEARCH

Open Access

The effectiveness of red-light therapy on myopia control depends on its direct effect: a mediation analysis

Qin Chen[1†], Hongyu Wei[2,3†], Zhi Da Soh[3†], Qin Zhu[4], Xian Shao[2], Cancan Xue[3], Hengtong Li[3], Yijin Tao[2], Min Hu[4], Ching-Yu Cheng[3,5,6,7] and Hua Zhong[2*]

Abstract

Purpose To investigate whether alterations in subfoveal choroidal thickness (SFCT) and peripheral retinal refraction mediate the efficacy of repeated low-level red-light (RLRL) therapy for myopia control.

Methods We conducted a mediation analysis within a multicenter, randomized controlled trial. A total of 300 myopic children were included in this analysis. Participants in RLRL group wore single-vision spectacles (SVSs) and received red-light therapy twice daily for 12 months. The control group wore SVSs only, without any additional myopia-control intervention. Axial length (AL), spherical equivalent refraction (SER), SFCT and total refractive difference value (TRDV) were measured at baseline and at follow-up visits over 12 months.

Results After 12 months, RLRL therapy significantly attenuated axial length and myopic shift in spherical equivalence compared with controls, accompanied by sustained increases in choroidal thickness and reductions in peripheral hyperopic defocus. Mediation analysis revealed that changes in choroidal thickness and peripheral refraction accounted for 24.83% and 6.79%, respectively, of the effect on axial length inhibition, and 28.79% and 5.97% of the effect on spherical equivalent control. The remaining effects were attributable to the direct impact of red-light (68.39% for axial length and 65.24% for spherical equivalence). Notably, choroidal thickening emerged as the predominant mediator within the first three months.

Conclusions RLRL therapy exerted a predominantly direct effect on myopia control, while changes in choroidal thickness and peripheral defocus partially mediated its effectiveness.

Trial registration This trial was registered at the Chinese Clinical Trial Registry on January 30, 2021, with trial registration number: ChiCTR2100042836. https://www.chictr.org.cn/showprojEN.html?proj=120971.

Keywords Repeated low-level red-light therapy, Myopia control, Mediation analysis, Choroidal thickness, Peripheral retinal refraction

[†]Qin Chen, Hongyu Wei and Zhi Da Soh contributed equally as first author.

*Correspondence:
Hua Zhong
zhoculist@163.com

Full list of author information is available at the end of the article

© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

Background

Myopia has surged into a global public-health crisis, affecting especially Asian youth [1]. In China, annual myopia incidence climbs from 7.8% in 6-year-olds to 25.3% 12-year-olds [2]. Beyond blurred vision, unchecked progression raises the risk of retinal detachment and myopic degeneration [3]. To combat this, the International Myopia Institute recommends optical, pharmacological, environmental, behavioral, and surgical approaches [4], among these options, repeated low-level red-light therapy has shown promise in slowing axial elongation and myopic shift [5, 6].

A growing body of work points to choroidal thickening as a key mediator, specifically, subfoveal thickening at three months strongly predicts long-term myopia control [7]. Similar choroidal responses have been observed following orthokeratology and low-dose atropine treatments [8, 9]. These findings align with the defocus theory, where myopic defocus signal induces a compensatory choroidal thickening to counter peripheral hyperopic and restrain axial elongation [10, 11].

Therefore, accurate peripheral refractive assessment is essential for understanding and optimizing control interventions. Multispectral Refraction Topography (MRT) offers repeatable measurements across a 53° fundus field [12] and has showed its value in quantifying orthokeratology-induced shifts in peripheral refraction [13]. However, there is a notable lack of evidence detailing peripheral refraction changes after red-light therapy.

Thus, the aim of this study based on the randomized clinical trial is to investigate whether red-light therapy exerts its myopia-slowing effect by direct exposure or mediated through changes in subfoveal choroidal thickness and peripheral refraction.

Methods

Study design

The study is a mediation analysis of the 12-month data from the Red-light Therapy for Clinical and Preclinical Myopia Children Study, a multicenter randomized controlled trial (RCT) registered in the Chinese Clinical Trials Registry (ChiCTR2100042836) in 2021, of which the details have been published elsewhere [14]. The full protocol is provided in Supplement 1. This trial was approved by the Second People's Hospital of Yunnan Province Ethics Committee (2020133) and adhered to the tenets of the Declaration of Helsinki. The participants' guardians provided written informed consent before enrollment in the RCT. This study was aimed to analyze the efficacy of red-light therapy on myopia control, thus, only the data of clinical myopies [15] (with – 6.00 D ≤ SER ≤ -0.50 D) had been analyzed.

Children in the intervention group received repeated low-level red-light (RLRL) therapy twice a day for 3 min, with a minimum of 4 h between sessions, 5 days per week, wearing single-vision spectacles (SVSs); the control group only wore SVSs. The RLRL intervention was provided by the Myopia Amblyopia Comprehensive Treatment Instrument (RS-200–2 A, Eyerising; wavelength: 650 ± 10 nm; light output power: 2.0 ± 0.5 mW, output power at a distance of 100 mm: 1.07- 1.42mW; illuminance approximately 1600 lx; laser spot diameter: 7 mm ± 3 mm, laser spot diameter at the observation port: 10 mm ± 2 mm).

Data collection

Axial length (AL) and cycloplegic refraction assessment

Before cycloplegia, visual acuity (GB/T 11533 – 2011) and slit lamp microscopy (Topcon, Japan) were performed. AL and keratometry (K) were measured with the IOL master (Carl Zeiss 500, Meditec). Five AL measurements with a difference of less than 0.05 mm were taken and averaged.

Cycloplegia was induced with Mydrin-P (Santen) eye drops, which were administered to both eyes at 0, 5, and 10 min. Pupil size and light reflex were observed 20 min after the final administration of eye drops. Cycloplegic autorefraction was performed with an autorefractor (Topcon, Japan), and three readings of each eye were taken and averaged until the desired precision (0.25 D for spherical and cylindrical power and 5 degrees for axis) was achieved. SER was calculated as the spherical value plus half the cylinder value.

OCT images and subfoveal choroidal thickness (SFCT) measurements

After cycloplegia, the "Enhanced HD Line" mode in OCT (Optovue, Fremont, CA, USA) was used to acquire choroidal tomography images. Choroidal thickness was measured manually by the same examiner from the outer edge of the highly reflective retinal pigment epithelial layer to the inner edge of the sclera under the macula fovea (eFigure 1 in Supplement 2). During the examination, eye-tracking and motion-correction was enabled. Scans exhibiting noticeable artifacts or a quality score less than 6 points (on a 10-point scale) were immediately repeated until a satisfactory image (≥ 6/10) was obtained. Participants were instructed to refrain from engaging in strenuous physical activities for at least one hour before the examinations and to avoid consuming caffeinated or other functional drinks, as these activities can impact choroidal thickness [16, 17].

MRT and retinal peripheral refraction measurements

The same examiner used Multispectral Refraction Topography (MRT; MSI C2000, Thondar, Shenzhen, China) to capture 53° retinal refraction in 2–3 s with enabled fixation control. The multispectral imaging

and lens compensation algorithms compute central and peripheral refractions to generate topographic maps. Foveal refraction was defined as the central refraction. The refraction difference value (RDV) at each eccentricity was its refraction minus the central refraction (eFigure 2 in Supplement 2), with positive values indicating hyperopic defocus and negative values indicating myopic defocus. The total refractive difference value (TRDV) was calculated automatically, reflecting the average retinal refractive difference of all eccentricities within 53° fundus field.

Statistical analysis

Right-eye data were used unless they failed quality checks, in which case left-eye data were substituted. Variables were tested for normality (Kolmogorov–Smirnov). Continuous data were presented as the mean ± SD and compared by independent t-test, while categorical data were expressed as number (percentage) and assessed with the Pearson's Chi-Square test. Linear mixed effects models (LMMs) were used to examine longitudinal changes in ocular parameters for each group and differences between groups. The longitudinal changes were set as the dependent variable. Group, time and the interaction of group × time were included as fixed variables. To eliminate the dependence of repeated measurements, random intercepts for participants were included. For pairwise comparisons and to obtain P-values, estimated marginal means (EMMs) were obtained from the fitted models. Pearson correlations assessed associations between parameters.

Mediation of red-light effects on axial length and spherical equivalent refraction by changes in subfoveal choroidal thickness and total refractive difference value was evaluated via structural equation models (SEM). Baseline characteristics were fitted in mediation models as confounding variables. The treatment (red-light group vs. control group) was modeled as an independent variable, and separate models were run for total (0–12 months) and interval (0–3, 3–6, 6–12 months) effects. The outcomes were changes in AL or SER during these intervals, with changs in SFCT and TRVD considered potential mediators. Path coefficients were estimated by maximum likelihood, with 95% CIs and p-values bootstrapped ($n = 1000$). Statistical analysis was conducted using R software (version 4.4.2), with significance levels set at $P < 0.05$ (two-sided).

Results

A total data of 300 eyes from 300 participants were analyzed (red-light group = 156 and control group = 144). The distribution of age, height, weight, K, AL, SER, SFCT and TRDV were well balanced at the baseline between the two groups. (eTable 1 in Supplement 2)

Longitudinal changes in ocular parameters

The longitudinal changes in axial length, spherical equivalent refraction, subfoveal choroidal thickness, and total refraction difference value over 12 months were shown in eTable 2 of Supplement 2 and Fig. 1. In the red-light group, after 3 months of red-light therapy, axial elongation was reduced (-0.04 ± 0.26 mm, $P = 0.088$) compared with baseline, spherical equivalent shifted hyperopically by 0.14 ± 0.40 D ($P < 0.001$), subfoveal choroid thickened by 19.54 ± 15.69 μm ($P < 0.001$), and peripheral refraction exhibited a small myopic shift (-0.05 ± 0.34 D, $P = 0.087$). Thereafter, axial length modestly increased, spherical equivalent gradually regressed toward emmetropia, while subfoveal choroid continued to thicken, reaching a maximum at 12 months (23.46 ± 19.62 μm, $P < 0.001$), and the peripheral hyperopic defocus steadily reduced, peaking at -0.21 ± 0.44 D at 12 months ($P < 0.001$).

By contrast, the control group exhibited continuous axial elongation, progressive myopic shift in spherical equivalent throughout the 12-month follow-up (all $P < 0.05$). However, the subfoveal choroidal thickness and peripheral refraction had no obvious change been observed during follow-up.

Between-group comparisons revealed a significant divergence in axial length, spherical equivalent and subfoveal choroidal thickness after 3 months (all $P < 0.001$), whereas divergence in peripheral refraction became significant after 6 months (all $P < 0.05$).

Correlation between ocular parameters and baseline characteristics over 12 months

The correlation matrix demonstrated the following significant correlations (all $p < 0.05$): In the red-light group, ΔAL (changes in axial length) demonstrated a moderate negative correlation with ΔSER (r = -0.38) and weak negative correlations with baseline AL (r = -0.27), baseline SER (r = -0.22), and age (r = -0.25); ΔSER (changes in spherical equivalent refraction) showed a weak positive correlation with ΔSFCT ($r = 0.24$); ΔTRDV (changes in total refraction difference value) exhibited a moderate negative correlation with its own baseline value (r = -0.45); Baseline AL was moderately negatively correlated with sex (r = -0.36) and weakly positively correlated with age ($r = 0.22$); Baseline SER showed a weak positive correlation with baseline SFCT ($r = 0.19$). In the control group, ΔAL demonstrated strong negative correlations with ΔSER (r = -0.66) and age (r = -0.54), as well as weak negative correlations with ΔSFCT (r = -0.21) and baseline AL (r = -0.21), ΔAL also showed a moderate positive correlation with ΔTRDV ($r = 0.33$), and a weak positive correlation with baseline TRDV ($r = 0.18$); ΔSER exhibited weak negative correlations with ΔTRDV (r = -0.24) and baseline SER (r = -0.21), a moderate positive correlation with age ($r = 0.36$), and weak positive correlations

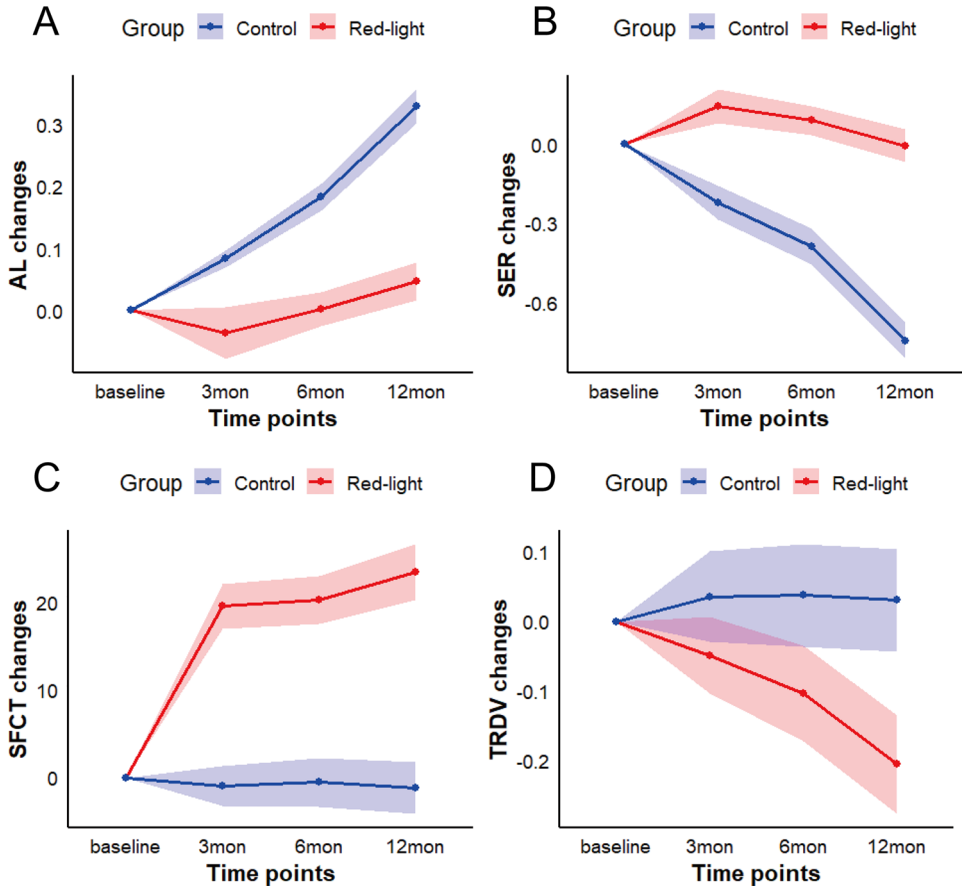


Fig. 1 Graph of longitudinal changes in ocular parameters. Longitudinal changes in the in axial length, spherical equivalent refraction, subfoveal choroidal thickness, and total refraction difference value during 12 months were presented graphically as mean ± 95% confidence intervals. (**A**) Longitudinal changes of axial length. (**B**) Longitudinal changes of spherical equivalent refraction. (**C**) Longitudinal changes of subfoveal choroidal thickness. (**D**) Longitudinal changes of total refraction difference value. Values on the x-axis correspond to baseline, 3 months, 6 months and 12 months, respectively

with ΔSFCT ($r = 0.18$) and baseline AL ($r = 0.23$); ΔTRDV showed weak negative correlations with its baseline value (r = -0.25) and baseline AL (r = -0.17); Baseline AL was strongly negatively correlated with baseline SER (r = -0.54) and moderately positively correlated with age ($r = 0.34$); Baseline SER demonstrated weak negative correlations with age (r = -0.30) and sex (r = -0.24). (Fig. 2)

Mediation analysis

During the whole stage (baseline to 12 months), ΔSFCT and ΔTRDV partially mediated the effect of red-light therapy on ΔAL, accounting for 24.82% and 6.79% of the total effect (Ps < 0.05), while the direct effect contributed 68.39% ($P < 0.001$). In the first 3 months, ΔSFCT mediated 30.79% of the total effect of red-light therapy on ΔAL ($P = 0.004$), and the direct effect of red-light accounted for 66.62% ($P = 0.045$), whereas ΔTRDV showed no significant mediation effect (2.59%, $P = 0.170$). For the subsequent periods (3-6months and 6–12 months), neither ΔSFCT nor ΔTRDV showed significant mediation effects, however, the direct effect of red-light therapy accounted 94.47% and 92.95% (Ps < 0.001), respectively. (Table 1; Fig. 3).

Similarly, ΔSFCT and ΔTRDV partially mediated the effect of red-light therapy on ΔSER during the whole period (baseline to 12-month), representing 28.79% and 5.97% of the total effect (Ps < 0.05), whereas the direct effect of red-light accounted for 65.24% ($P < 0.001$). In the first 3 months, ΔSFCT mediated 30.39% of the total effect of red-light therapy on ΔSER ($P < 0.001$), and the direct effect of red-light accounted for 67.81% ($P < 0.001$), whereas ΔTRDV did not show a significant mediating effect (1.80%, $P = 0.293$). Neither ΔSFCT nor ΔTRDV showed significant mediation effects for succeeding periods, but the direct effect of red-light therapy represented 99.76% and 90.11% (Ps < 0.001) of total effect, respectively (Table 2; Fig. 3).

Discussion

Our findings indicate that the 12 months of twice daily red-light therapy markedly reduced axial elongation and myopic shifts in spherical equivalence along with

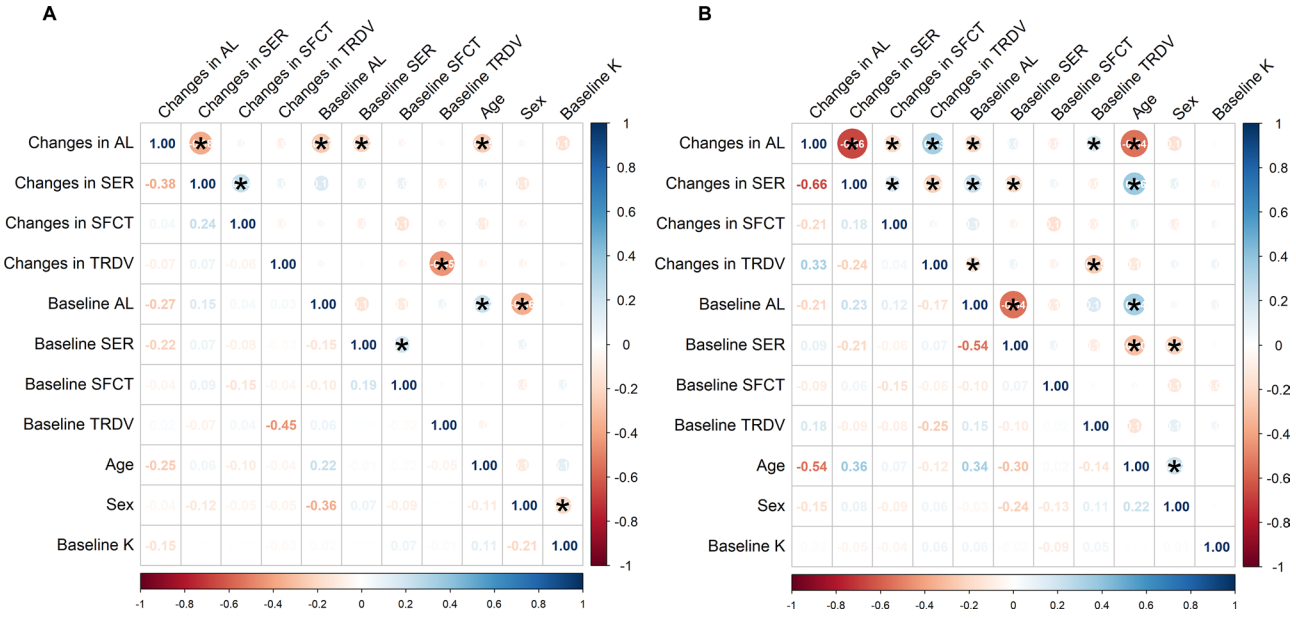


Fig. 2 Graphical display of correlations between changes in ocular parameters and baseline characteristics within red-light group (**A**) and control group (**B**). Correlation coefficients were shown in the lower triangle (the clearer the font, the higher the correlation coefficient). In the upper triangle, the same values were shown as colored bubbles. Red bubbles indicate negative values. Blue bubbles indicate positive values. The lighter and smaller the bubbles, the closer the values were to zero. Significance coefficients were indicated by black stars in the upper triangle (significance level fixed at $P < 0.05$)

a sustained increase in subfoveal choroidal thickness and decrease in peripheral hyperopic defocus. These effects suggest that red-light therapy may offer a multi-dimensional protection against myopia development and advancement.

Although prior studies have linked red-light related myopia control to choroidal thickening [7, 18, 19], it remains unclear: (1) How does red-light therapy alter peripheral refraction? (2) What drives its choroidal thickening? (3) To what extent does choroidal thickening mediate myopia control? We are the first to show a reduction in hyperopic defocus in the 53° fundus view with red-light therapy over 12 months. Mediation analysis revealed that choroidal thickening contributes partially to myopia control, especially in the early stage, though the primary benefit arises directly from red-light exposure.

Although concerns have been raised about possible cone or retina damage [20, 21], we observed no serious adverse events or fundus abnormalities. A recent review of 20 studies likewise found no irreversible visual or structural harm with repeated low-level red-light therapy [22]. Given ongoing safety debates, elucidating the underlying mechanisms of red-light's protective effects remains essential for its future clinical adoption.

Response of choroidal thickness and retinal peripheral refraction of red-light therapy

Sustained choroidal thickening with red-light therapy over entire course of therapy has been observed in many clinical trials, and in the early stages of red-light use, the choroid grows more vigorously [7, 23, 24], and the increase in choroidal thickness after red-light therapy may be related to an increase in choroidal blood flow [18, 25]. We found that the administration of red-light therapy increased subfoveal choroidal thickness, especially in the early stages, but levels stabilized after three months. However, a study indicated that choroidal thickening in children was only observed during red-light treatment, and that the SFCT returned to baseline levels three months after treatment cessation [23].

Increasing studies have provided evidence for the effects of myopia-control interventions on decreasing peripheral hyperopic defocus [26, 27], and our study provides new evidence for the effects of red-light therapy on peripheral refraction and reports that the red-light therapy mildly reduced the peripheral hyperopic defocus compared with baseline in the total 53° of fundus field. A study on animals backs our findings: long-wavelength narrow-band light inhibited axial elongation by inducing myopic defocus, and red-light-induced alterations in refraction were linked to reduced vitreous chamber elongation and thickened choroid [28]. However, compared with other interventions (e.g. multifocal spectacles and orthokeratology), red-light therapy has no such significant effect on shifting peripheral retinal defocus in our study [26, 27]. In other words, red-light therapy appears to exert its myopia control effects primarily through non-optical mechanisms. This suggests that it may serve as a complementary intervention, particularly for children

Table 1 Path coefficient for linking treatment to changes in AL

Pathway	Unstandardized[†] Coefficients	Standardized[‡] Coefficients	*P* Value	% Mediated[§]
	β (95% CI)	β (95% CI)		
Total stage (0–12 months)				
Indirect effects				
Red-light→ △SFCT→△AL	-0.09 (-0.11 to -0.06)	-0.23 (-0.30 to -0.17)	**< 0.001***	24.82
Red-light→ △TRDV→△AL	-0.02 (-0.05 to -0.01)	-0.06 (-0.12 to -0.01)	**0.015***	6.79
Direct effect				
Red-light→ △AL	-0.24 (-0.27 to -0.20)	-0.65 (-0.70 to -0.59)	**< 0.001***	68.39
Total effect				
Red-light→ △AL	-0.35 (-0.39 to -0.30)	-0.95 (-1.04 to -0.85)	**< 0.001***	100
Stage 1 (0–3 months)				
Indirect effects				
Red-light→ △SFCT→△AL	-0.02 (-0.02 to -0.01)	-0.14 (-0.24 to -0.05)	**0.004***	30.79
Red-light→ △TRDV→△AL	0.00 (0.00 to 0.00)	-0.01 (-0.03 to 0.01)	0.170	2.59
Direct effect				
Red-light→ △AL	-0.04 (-0.05 to -0.02)	-0.30 (-0.60 to -0.01)	**0.045***	66.62
Total effect				
Red-light→ △AL	-0.06 (-0.07 to -0.04)	-0.46 (-0.77 to -0.15)	**0.004***	100
Stage 2 (3–6 months)				
Indirect effects				
Red-light→ △SFCT→△AL	0.00 (0.00 to 0.00)	0.00 (-0.02 to 0.01)	0.541	3.16
Red-light→ △TRDV→△AL	0.00 (0.00 to 0.00)	0.00 (0.00 to 0.01)	0.400	2.37
Direct effect				
Red-light→ △AL	-0.01 (-0.03 to 0.00)	-0.14 (-0.20 to -0.08)	**< 0.001***	94.47
Total effect				
Red-light→ △AL	-0.01 (-0.03 to -0.00)	-0.15 (-0.21 to -0.08)	**< 0.001***	100
Stage3 (6–12 months)				
Indirect effects				
Red-light→ △SFCT→△AL	0.00 (-0.01 to 0.00)	-0.02 (-0.04 to 0.01)	0.168	3.43
Red-light→ △TRDV→△AL	0.00 (-0.01 to 0.00)	-0.02 (-0.04 to 0.01)	0.203	3.62
Direct effect				
Red-light→ △AL	-0.07 (-0.09 to -0.06)	-0.41 (-0.50 to -0.32)	**< 0.001***	92.95
Total effect				
Red-light→ △AL	-0.08 (-0.10 to -0.06)	-0.44 (-0.53 to -0.35)	**< 0.001***	100

Path coefficient estimates were obtained using maximum likelihood estimation with 95% confidence intervals and P values were derived from bootstrapping with 1000 bootstrap replicates within a structural equation modelling framework. P values of unstandardized coefficients were valid for standardized coefficients. Age, baseline AL, SFCT and TRDV were fitted in models as confounding variables

* Statistically significant ($P < 0.05$)

[†] Unstandardized effect in raw outcome units

[‡] Standardized effect in standard deviation units of outcome

[§] % Mediated = specific effect coefficient / total effect coefficient

whose myopia continues to progress despite conventional optical treatments such as orthokeratology or multifocal lenses.

Choroidal thickening was evident during the early phase of red-light exposure, likely contributing to axial length shortening by anteriorly shifting the retinal plane. However, this thickening response was sustained indefinitely; a gradual plateau in choroidal growth was observed after the initial three months, consistent with findings from previous studies [5, 7]. Mediation analysis further indicated that the effect of choroidal thickening was most pronounced during the early stage, yet it also contributed significantly to the overall treatment period, suggesting both an early and sustained role in myopia control. In contrast, inter-group differences in TRDV emerged only after six months of red-light exposure, and its mediating effect on myopia control was significant only over the 12-month period, this suggests that changes in retinal peripheral refraction reflect a slower, long-term structural adaptation may be essential for maintaining the therapeutic effect of red-light therapy over time.

More research is in support of the change in choroidal thickness as a compensatory change in the retina receiving defocused signals [27], the causal relationship between choroidal thickness and the peripheral defocus

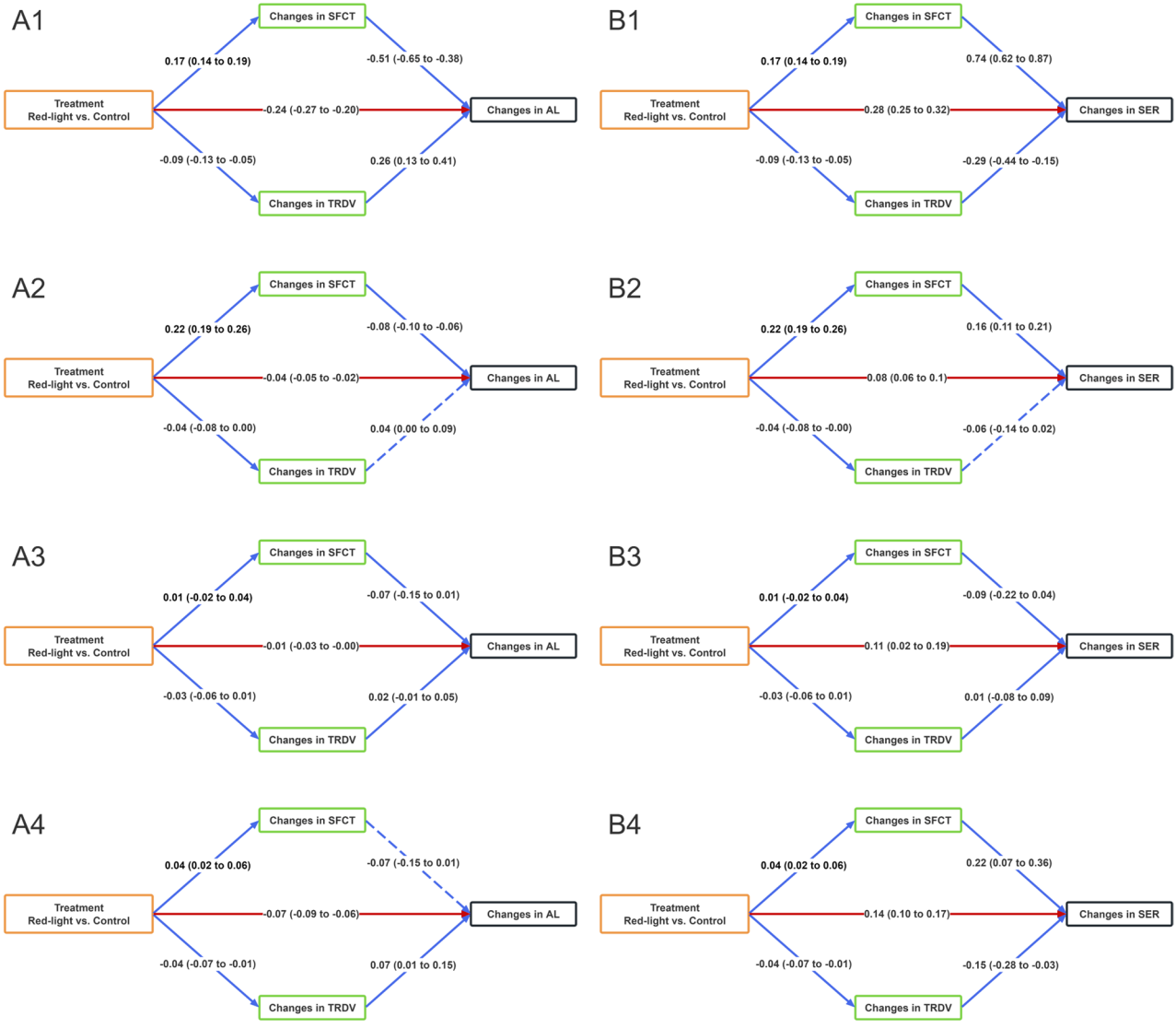


Fig. 3 Pathways of red-light therapy in myopia control. Variables are connected by arrows representing a path in each model shown. Treatment exerts effects indirectly (blue arrows) via mediators or directly (red arrows). Unstandardized pathway coefficients are shown as β (95% CIs), with bold arrows indicating statistically significant effects ($P < 0.05$), with dotted arrows indicating statistically insignificant effects ($P \geq 0.05$). Fig A reflects the effect on AL elongation, and Fig B reflects the effect on SER progression: (**A1**) and (**B1**) show the changes from baseline to 12 months; (**A2**) and (**B2**) show the changes from baseline to 3 months; (**A3**) and (**B3**) show the changes from 3 to 6 months; (**A4**) and (**B4**) show the changes from 6 to 12 months

state has not yet been determined and more research is needed to verify the specific pathways.

Relationship between the choroidal thickness and myopia progression

The choroid plays a critical role in the development of myopia. An increasing number of studies have found that choroidal thickness correlates with both spherical equivalent refraction and axial length and suggested that choroidal thickness might be considered an accurate indicator for evaluating and predicting myopia progression [29, 30]. In our study, we showed that choroidal thickening could explain part of the effect of myopia control during the early period. It was supported by other studies with different interventions [7, 9], the changes in choroidal thickness could serve as valuable biomarkers for assessing myopia progression and guiding therapeutic interventions in the early stages of treatment.

However, it should be noted that although choroidal thickness shows some relationship with axial length and spherical equivalent refraction, changes in choroidal thickness alone are not sufficient to explain changes in spherical equivalent refraction or axial length [31, 32], including age and genetic predisposition, which play an important role in determining myopia. While there is strong evidence linking choroidal thickening to myopia

Table 2 Path coefficient for linking treatment to changes in SER

Pathway	Unstandardized[†] Coefficients	Standardized[‡] Coefficients	*P* Value	% Mediated[§]
	β (95% CI)	β (95% CI)		
Total stage (0–12 months)				
Indirect effects				
Red-light→ △SFCT→ △SER	0.12 (0.10 to 0.15)	0.33 (0.27 to 0.39)	**< 0.001***	28.79
Red-light→ △TRDV→ △SER	0.03 (0.01 to 0.05)	0.06 (0.02 to 0.10)	**0.016***	5.97
Direct effect				
Red-light→ △SER	0.28 (0.25 to 0.32)	0.71 (0.66 to 0.75)	**< 0.001***	65.24
Total effect				
Red-light→ △SER	0.43 (0.38 to 0.48)	1.09 (0.99 to 1.18)	**< 0.001***	100.00
Stage 1 (0–3 months)				
Indirect effects				
Red-light→ △SFCT→ △SER	0.04 (0.02 to 0.05)	0.20 (0.11 to 0.29)	**< 0.001***	30.39
Red-light→ △TRDV→ △SER	0.00 (0.00 to 0.01)	0.01 (-0.01 to 0.03)	0.293	1.80
Direct effect				
Red-light→ △SER	0.08 (0.06 to 0.10)	0.44 (0.35 to 0.54)	**< 0.001***	67.81
Total effect				
Red-light→ △SER	0.12 (0.09 to 0.14)	0.65 (0.52 to 0.78)	**< 0.001***	100.00
Stage 2 (3–6 months)				
Indirect effects				
Red-light→ △SFCT→ △SER	0.01 (-0.01 to 0.01)	0.00 (-0.01 to 0.01)	0.816	0.52
Red-light→ △TRDV→ △SER	0.00 (-0.01 to 0.01)	0.00 (-0.04 to 0.01)	0.909	-0.28
Direct effect				
Red-light→ △SER	0.11 (0.02 to 0.19)	0.16 (0.02 to 0.29)	**0.024***	99.76
Total effect				
Red-light→ △SER	0.11 (0.02 to 0.19)	0.16 (0.02 to 0.29)	**0.024***	100.00
Stage3 (6–12 months)				
Indirect effects				
Red-light→ △SFCT→ △SER	0.01 (0.00 to 0.02)	0.03 (0.00 to 0.05)	**0.044***	5.90
Red-light→ △TRDV→ △SER	0.01 (0.00 to 0.02)	0.02 (-0.04 to 0.04)	0.159	3.99
Direct effect				
Red-light→ △SER	0.14 (0.10 to 0.17)	0.42 (0.33 to 0.50)	**< 0.001***	90.11
Total effect				
Red-light→ △SER	0.15 (0.12 to 0.18)	0.46 (0.37 to 0.56)	**< 0.001***	100.00

Path coefficient estimates were obtained using maximum likelihood estimation with 95% confidence intervals and P values were derived from bootstrapping with 1000 bootstrap replicates within a structural equation modelling framework. P values of unstandardized coefficients were valid for standardized coefficients. Age, baseline SER, SFCT and TRDV were fitted in models as confounding variables

* Statistically significant ($P < 0.05$)

† Unstandardized effect in raw outcome units

‡ Standardized effect in standard deviation units of outcome

§ % Mediated = specific effect coefficient / total effect coefficient

control, more research is needed to fully understand the mechanisms involved.

Relationship between the peripheral retinal defocus and myopia progression

The peripheral retina is thought to play a critical role in the regulation of eye growth and refractive state [33]. Animal model suggested that retinal defocus trigger a signaling cascade that causes changes to the retinal pigment epithelium, chemical signals transmitted through the choroid cause changes in scleral extracellular matrix synthesis, adjusting the ocular growth [34]. According to the theory of peripheral retinal defocus, hyperopic defocus induces axial elongation, whereas myopic defocus slows or even reverses it [10, 11]. Although peripheral retinal refraction did not shift from hyperopic defocus at baseline to myopic defocus at 12-month in our study, mediation analysis suggests that the reduction in peripheral hyperopic defocus still contributes to myopia control. Therefore, further validation is warranted to determine whether longer-term therapy induces a shift of peripheral retinal refraction to myopic defocus. Additionally, the rebound effect on peripheral retinal refraction following therapy cessation remains to be investigated.

Similar to the effect of choroidal thickening in myopia control, the effect of peripheral retinal refraction alone

is not sufficient to explain the effect of red-light therapy on myopia control. The size and location of the peripheral defocus zone are crucial for effective myopia control, with larger treatment zones providing stronger effects [35]. Further studies are needed to investigate the specific amount and location of defocus after red-light therapy, and to explore the mechanisms behind variations in peripheral defocus.

The potential mechanisms of red-light direct effect on myopia control

Thought the mediation analysis, we can find that no matter what time period in, the myopia-controlling effect of red-light therapy was primarily driven by its direct effect. The possible mechanisms of RLRL therapy on myopia control had been described in a review that published by our team [36], including: improvement of mitochondrial function and bioenergetics, enhancement of neuroprotection, regulation of oxidative stress and inflammation, dopamine release and nitric oxide (NO) production, choroidal thickening and scleral remodeling. One potential mechanism of choroidal thickening is that red-light stimulates the synthesis and release of dopamine (DA) by the retina [37], and DA may promote choroidal thickening by releasing nitric oxide (NO), a vasodilator to improve choroidal blood flow [38–40]. Beyond vascular effects, emerging evidence suggests that red-light exposure may also influence scleral remodeling by modulating collagen synthesis and extracellular matrix turnover [41, 42], potentially through mitochondrial activation and redox-sensitive signaling pathways [43], both of which are implicated in ocular structural regulation. Additionally, recent studies have identified retinal oxytocin signaling as a potential contributor to ocular growth control. Oxytocin, a neuropeptide with anti-inflammatory and vasoregulatory properties, has been detected in retinal tissue and may interact with dopaminergic and NO-mediated pathways [44]. Although its role in red-light therapy remains speculative, it represents a promising area for future mechanistic research.

Together, these pathways support a biologically plausible framework for the direct effects of RLRL therapy, involving both vascular and structural remodeling processes. Further experimental studies are warranted to validate these mechanisms and clarify their relevance in clinical myopia control.

Overall, many studies have confirmed the effectiveness of low-level red-light treatment in myopia control and prevention, which cannot be explained by choroidal thickening or myopic defocus of the peripheral retina alone. To enhance the generalizability and clinical relevance of our findings, future studies should include multi-ethnic populations in order to account for potential variability in treatment response arising from genetic and environmental differences. Expanding the age range to include both younger children and older adolescents will help to clarify the effect of RLRL therapy on different age group. Furthermore, evaluating multiple RLRL devices and treatment protocols is essential to assess consistency across platforms and to optimize therapeutic parameters. Such investigations will be crucial in validating and extending our current results, and will ultimately support broader clinical translation of RLRL therapy for myopia control.

Strengths and limitations

To the best of our knowledge, our study is the first to report on the changes in human peripheral retinal refraction after red-light treatment. Additionally, we assessed the mediation effect of the alterations in choroidal thickness and peripheral retinal refraction on the efficacy of myopia control under the red-light therapy.

There are some limitations in this study. Firstly, subfoveal choroidal thickness was measured manually, despite we tried to reduce bias by using the same examiner, the absence of inter-observer repeatability assessment may still render the measurements susceptible to potential bias. This result can to be future validated using automated techniques or devices. Secondly, the follow-up period was short, and this study remains a single study. Further verification of the mediation effects is required in larger, more diverse populations of red-light therapy cohorts over an extended time period. In addition, further validation of the mediation effects is warranted, including consideration of supplementary mediators, potential unmeasured confounding, and sensitivity analyses to assess the robustness of the findings. Thirdly, the location of the peripheral defocus zone is critical for evaluating the effects of myopia control. The peripheral retinal defocus at different locations should be investigated further. Finally, we propose that the primary effect of red-light therapy on myopia control is likely its' direct effect. However, the underlying mechanism, methodological framework, and interpretation remain unclear. To address these gaps, further experiments will be conducted to elucidate the mechanism and refine the methodological approach.

Conclusions

Repeated low-level red-light therapy significantly reduced the myopic progression in terms of axial length and spherical equivalent refraction and caused a sustained increase in subfoveal choroidal thickness and decrease in peripheral hyperopic defocus over 12 months. The effectiveness of red-light therapy on myopia control is mainly due to its direct impact, whereas choroidal thickening and the reduction of hyperopic peripheral defocus only partially explain its effectiveness.

Abbreviations

AL	Axial length
CT	Choroidal thickness
EMMs	Estimated marginal mean
LMMs	Linear mixed effects models
MRT	Multispectral Refraction Topography
RCT	Randomized controlled trial
RLRL	Repeated low-level red-light
SER	Spherical equivalent refraction
SEM	Structural equation model
SFCT	Subfoveal choroidal thickness
SVSs	Single-vision spectacles
TRDV	Total refractive difference value
UCVA	Uncorrected visual acuity

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12967-025-07514-y.

Supplementary Material 1

Supplementary Material 2

Author contributions

Qin Chen: Supervision, Provide materials and resources, Writing – review & editing; Hongyu Wei: Data curation, Statistical analysis, Data visualization, Writing – original draft; Zhi Da Soh: Data visualization, Writing – review & editing; Qin Zhu: Data curation; Xian Shao: Data curation; Cancan Xue: Statistical analysis, Writing – review & editing; Hengtong Li: Statistical analysis; Yijin Tao: Writing – review & editing; Min Hu: Provide materials and resources; Ching-Yu Cheng: Writing – review & editing; Hua Zhong: Supervision, Provide materials and resources, Writing – review & editing, Funding acquisition.

Funding

This study was supported by the National Natural Science Foundation of China (Grant No. 82360212) and the Yunnan Fundamental Research Kunming Medical University Projects (Grant No. 202301AY070001-002), the First-Class Discipline Team of Kunming Medical University Project (Grant No. 2024XKTDTS01) and the Key Clinical Specialty of Ophthalmology in Yunnan Province Project (Grant No. 300070).

Data availability

The datasets generated and analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

The study was approved by the Second People's Hospital of Yunnan Province Ethics Committee (Grant No. 2020133) and adhered to the tenets of the Declaration of Helsinki. The participants' guardians provided written informed consent prior to enrollment in the RCT. This trial was conducted in accordance with the Consolidated Standards of Reporting Trials (CONSORT) reporting guidelines.

Consent for publication

Not applicable.

Competing interests

The authors have no proprietary or commercial interest in any materials discussed in this article.

Author details

[1]Department of Ophthalmology, The First Affiliated Hospital with Nanjing Medical University, 300 Guangzhou Road, Nanjing, Jiangsu 210029, China
[2]Department of Ophthalmology, The First Affiliated Hospital of Kunming Medical University, 295 Xichang Road, Kunming, Yunnan 650032, China
[3]Department of Ocular Epidemiology, Singapore Eye Research Institute, 20 College Road, Singapore 169856, Singapore
[4]Department of Pediatric Ophthalmology, Affiliated Hospital of Yunnan University, 176 Qingnian Road, Kunming, Yunnan 650021, China
[5]Ophthalmology & Visual Sciences Academic Clinical Program, Duke-NUS Medical School, 8 College Road, Singapore 169857, Singapore
[6]Centre for Innovation and Precision Eye Health, National University of Singapore, 1E Kent Ridge Road, NUHS Tower Block, Level 7, Singapore 119228, Singapore
[7]Department of Ophthalmology, National University of Singapore, 1E Kent Ridge Road, NUHS Tower Block, Level 7, Singapore 119228, Singapore

Received: 30 July 2025 / Accepted: 23 November 2025
Published online: 02 December 2025

References

1. Pan CW, Dirani M, Cheng CY, Wong TY, Saw SM. The age-specific prevalence of myopia in Asia: a meta-analysis. Optom Vis Sci. 2015;92(3):258–66. https://doi.org/10.1097/OPX.0000000000000516.
2. Li SM, Wei S, Atchison DA, et al. Annual incidences and progressions of myopia and high myopia in Chinese schoolchildren based on a 5-Year cohort study. Invest Ophthalmol Vis Sci. 2022;63(1):8. https://doi.org/10.1167/iovs.63.1.8.
3. Lee SSY, Lingham G, Sanfilippo PG, et al. Incidence and progression of myopia in early adulthood. JAMA Ophthalmol. 2022;140(2):162–9. https://doi.org/10.1001/jamaophthalmol.2021.5067.
4. Wildsoet CF, Chia A, Cho P, et al. IMI - interventions myopia institute: interventions for controlling myopia onset and progression report. Invest Ophthalmol Vis Sci. 2019;60(3):M106–31. https://doi.org/10.1167/iovs.18-25958.
5. He X, Wang J, Zhu Z, et al. Effect of repeated Low-level red light on myopia prevention among children in China with premyopia: a randomized clinical trial. JAMA Netw Open. 2023;6(4):e239612. https://doi.org/10.1001/jamanetworkopen.2023.9612.
6. Jiang Y, Zhu Z, Tan X, et al. Effect of repeated low-level red-light therapy for myopia control in children: a multicenter randomized controlled trial. Ophthalmology. 2022;129(5):509–19. https://doi.org/10.1016/j.ophtha.2021.11.023.
7. Xiong R, Zhu Z, Jiang Y, et al. Longitudinal changes and predictive value of choroidal thickness for myopia control after repeated Low-Level Red-Light therapy. Ophthalmology. 2023;130(3):286–96. https://doi.org/10.1016/j.ophtha.2022.10.002.
8. Li Z, Hu Y, Cui D, Long W, He M, Yang X. Change in subfoveal choroidal thickness secondary to orthokeratology and its cessation: a predictor for the change in axial length. Acta Ophthalmol. 2019;97(3):e454–9. https://doi.org/10.1111/aos.13866.
9. Yam JC, Jiang Y, Lee J, et al. The association of choroidal thickening by Atropine with treatment effects for myopia: Two-Year clinical trial of the Low-concentration Atropine for myopia progression (LAMP) study. Am J Ophthalmol. 2022;237:130–8. https://doi.org/10.1016/j.ajo.2021.12.014.
10. Wallman J, Winawer J. Homeostasis of eye growth and the question of myopia. Neuron. 2004;43(4):447–68. https://doi.org/10.1016/j.neuron.2004.08.008.
11. Chiang STH, Phillips JR, Backhouse S. Effect of retinal image defocus on the thickness of the human choroid. Ophthalmic Physiol Opt. 2015;35(4):405–13. https://doi.org/10.1111/opo.12218.
12. L W, J R, D W, et al. Agreement and repeatability of central and peripheral refraction by one novel multispectral-based refractor. Front Med. 2021;8. https://doi.org/10.3389/fmed.2021.777685.
13. Wen-Ting Tang XNL, Wen-Ting Tang XNL. One-year results for myopia control of orthokeratology with different back optic zone diameters: a randomized trial using a novel multispectral-based topographer. Qwer. 2024;17(2):324–30. https://doi.org/10.18240/ijo.2024.02.15.
14. Wei H, Thakur S, Zhu Q, et al. Effects on choroid after repeated Low-Level Red-Light therapy in preclinical and clinical myopic children: a randomized controlled Trial. Photodiagnosis Photodyn Ther. Published Online May. 2025;29:104651. https://doi.org/10.1016/j.pdpdt.2025.104651.
15. Flitcroft DI, He M, Jonas JB, et al. IMI – Defining and classifying myopia: A proposed set of standards for clinical and epidemiologic studies. Invest Ophthalmol Vis Sci. 2019;60(3):M20–30. https://doi.org/10.1167/iovs.18-25957.

16. Yeung S, You Y, Howe K, Yan P. Choroidal thickness in patients with cardiovascular disease: A review. Surv Ophthalmol. 2020;65(4):473–86. https://doi.org/10.1016/j.survophthal.2019.12.007.
17. Altinkaynak H, Ceylan E, Kartal B, Keleş S, Ekinci M, Olcaysu OO. Measurement of choroidal thickness following caffeine intake in healthy subjects. Curr Eye Res. 2016;41(5):708–14. https://doi.org/10.3109/02713683.2015.1020168.
18. Xuan M, Zhu Z, Jiang Y, et al. Longitudinal changes in choroidal structure following repeated low-level red-light therapy for myopia control: secondary analysis of a randomized controlled trial. Asia Pac J Ophthalmol (Phila). 2023. https://doi.org/10.1097/APO.0000000000000618.
19. Wang H, Zhong H, Zhang J, Wei W, Cui X, Zheng W. Impact of repeated low-level red-light exposure on choroidal thickness and blood flow in pediatric patients: A SS-OCTA study. Photodiagnosis Photodyn Ther. 2024;50:104412. https://doi.org/10.1016/j.pdpdt.2024.104412.
20. Liao X, Yu J, Fan Y, et al. Cone density changes after repeated Low-Level red light treatment in children with myopia. JAMA Ophthalmol Published Online April. 2025;24:e250835. https://doi.org/10.1001/jamaophthalmol.2025.0835.
21. Liu H, Yang Y, Guo J, Peng J, Zhao P. Retinal damage after repeated Low-level Red-Light laser exposure. JAMA Ophthalmol. 2023;141(7):693–5. https://doi.org/10.1001/jamaophthalmol.2023.1548.
22. Chen Y, Xiong R, Yang S, et al. Safety of repeated low-level red-light therapy for myopia: A systematic review. Asia Pac J Ophthalmol (Phila). 2024;13(6):100124. https://doi.org/10.1016/j.apjo.2024.100124.
23. Chen H, Wang W, Liao Y, et al. Low-intensity red-light therapy in slowing myopic progression and the rebound effect after its cessation in Chinese children: a randomized controlled trial. Graefes Arch Clin Exp Ophthalmol. 2023;261(2):575–84. https://doi.org/10.1007/s00417-022-05794-4.
24. Xiang K, Wang J, Zhu Z, et al. Changes in choroidal thickness in pre-myopic children after repeated low-level red-light therapy and their role in predicting myopia prevention and controlling myopic shift. Asia Pac J Ophthalmol (Phila). 2024:100115. https://doi.org/10.1016/j.apjo.2024.100115.
25. Liu G, Li B, Rong H, et al. Axial length shortening and choroid thickening in myopic adults treated with repeated Low-Level red light. J Clin Med. 2022;11(24):7498. https://doi.org/10.3390/jcm11247498.
26. Queirós A, Pinheiro I, Fernandes P. Peripheral defocus in orthokeratology myopia correction: systematic review and Meta-Analysis. J Clin Med. 2025;14(3):662. https://doi.org/10.3390/jcm14030662.
27. Alanazi M, Caroline P, Alshamrani A, Liu M. Impact of multifocal gas-permeable lens designs on short-term choroidal response, axial length, and retinal defocus profile. Int J Ophthalmol. 2024;17(2):247–56. https://doi.org/10.18240/ijo.2024.02.04.
28. Hung LF, Arumugam B, She Z, Ostrin L, Smith EL. Narrow-band, long-wavelength lighting promotes hyperopia and retards vision-induced myopia in infant rhesus monkeys. Exp Eye Res. 2018;176:147–60. https://doi.org/10.1016/j.exer.2018.07.004.
29. Zhu H, Liu C, Gao M, Zhang S, Zhang L, Zhao Q. Choroidal thickness in relation to diopter and axial length among myopic children. Front Med. 2023;10. https://doi.org/10.3389/fmed.2023.1241352.
30. Hansen MH, Kessel L, Li XQ, Skovgaard AM, Larsen M, Munch IC. Axial length change and its relationship with baseline choroidal thickness – a five-year longitudinal study in Danish adolescents: the CCC2000 eye study. BMC Ophthalmol. 2020;20(1):152. https://doi.org/10.1186/s12886-020-01427-8.
31. Ulaganathan S, Read SA, Collins MJ, Vincent SJ. Daily axial length and choroidal thickness variations in young adults: associations with light exposure and longitudinal axial length and choroid changes. Exp Eye Res. 2019;189:107850. https://doi.org/10.1016/j.exer.2019.107850.
32. Michalewski J, Michalewska Z, Nawrocka Z, Bednarski M, Nawrocki J. Correlation of choroidal thickness and volume measurements with axial length and age using swept source optical coherence tomography and optical low-coherence reflectometry. Biomed Res Int. 2014;2014:639160. https://doi.org/10.1155/2014/639160.
33. Smith EL, Ramamirtham R, Qiao-Grider Y, et al. Effects of foveal ablation on emmetropization and form-deprivation myopia. Invest Ophthalmol Vis Sci. 2007;48(9):3914–22. https://doi.org/10.1167/iovs.06-1264.
34. Rymer J, Wildsoet CF. The role of the retinal pigment epithelium in eye growth regulation and myopia: a review. Vis Neurosci. 2005;22(3):251–61. https://doi.org/10.1017/S0952523805223015.
35. Benavente-Pérez A, Nour A, Troilo D. Axial eye growth and refractive error development can be modified by exposing the peripheral retina to relative myopic or hyperopic defocus. Invest Ophthalmol Vis Sci. 2014;55(10):6765–73. https://doi.org/10.1167/iovs.14-14524.
36. Zhu Q, Cao X, Zhang Y, et al. Repeated Low-Level Red-Light therapy for controlling onset and progression of Myopia-a review. Int J Med Sci. 2023;20(10):1363–76. https://doi.org/10.7150/ijms.85746.
37. Cohen Y, Peleg E, Belkin M, Polat U, Solomon AS. Ambient illuminance, retinal dopamine release and refractive development in chicks. Exp Eye Res. 2012;103:33–40. https://doi.org/10.1016/j.exer.2012.08.004.
38. Nickla DL, Damyanova P, Lytle G. Inhibiting the neuronal isoform of nitric oxide synthase has similar effects on the compensatory choroidal and axial responses to myopic defocus in chicks as does the non-specific inhibitor L-NAME. Exp Eye Res. 2009;88(6):1092–9. https://doi.org/10.1016/j.exer.2009.01.012.
39. Nickla DL, Wildsoet CF. The effect of the nonspecific nitric oxide synthase inhibitor NG-nitro-L-arginine Methyl ester on the choroidal compensatory response to myopic defocus in chickens. Optom Vis Sci. 2004;81(2):111–8. https://doi.org/10.1097/00006324-200402000-00009.
40. Sekaran S, Cunningham J, Neal MJ, Hartell NA, Djamgoz MBA. Nitric oxide release is induced by dopamine during illumination of the carp retina: serial neurochemical control of light adaptation. Eur J Neurosci. 2005;21(8):2199–208. https://doi.org/10.1111/j.1460-9568.2005.04051.x.
41. Xiao Q, Zhang X, Chen ZL, Zou YY, Tang CF. An Evidence-Based narrative review of scleral hypoxia theory in myopia: from mechanisms to treatments. Int J Mol Sci. 2025;26(1):332. https://doi.org/10.3390/ijms26010332.
42. Mokoena DR, Houreld NN, Dhilip Kumar SS, Abrahamse H. Photobiomodulation at 660 Nm stimulates fibroblast differentiation. Lasers Surg Med. 2020;52(7):671–81. https://doi.org/10.1002/lsm.23204.
43. Yang J, Ouyang X, Fu H, et al. Advances in biomedical study of the myopia-related signaling pathways and mechanisms. Biomed Pharmacother. 2022;145:112472. https://doi.org/10.1016/j.biopha.2021.112472.
44. Lykke L, Ernst C, Bek T. The vasoactive effects of bradykinin, vasoactive intestinal peptide, calcitonin gene-related peptide and neuropeptide Y depend on the perivascular tissue in Porcine retinal arterioles in vitro. Acta Ophthalmol. 2024;102(3):349–56. https://doi.org/10.1111/aos.15742.

Publisher's note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.